use changes in gravity to mimic the effects of dark energy will look very different in the combined data, Hložek says. “Basically, I think we’ll run out of places to hide.”

What other cool things can you see if you have images of billions of galaxies? A big one is strong gravitational lensing, in which a galaxy or cluster of galaxies is distorting the space it’s in so much that the light from an object directly behind it gets split into multiple images, or spread out as an arc of light encircling it. Think of looking at a candle through the base of an empty wineglass—the curved glass spreads the light out in broad arcs or a circle instead of showing it to you as a single flame. When a gravitational lens does this, the individual images follow different paths through the distorted space. That means that if, for example, a supernova goes off in the lensed galaxy, it might be seen in one of the images *before* it shows up in another, because the light making up the second image took a longer path to get to us.

Aside from being a fabulous party trick,[fn3] time delay measurements like this give us a new way of measuring the expansion rate of the universe, since the distances involved are so large that the expansion becomes an important factor in the calculation. And we desperately need new ways to measure the expansion rate, because our current methods are giving us weirdly different answers.

As you’ll recall from Chapter 5, measuring the expansion rate (also known as the Hubble Constant) using supernovae gives us one number, and measuring it via the CMB gives us another. A slew of other measurements have failed to resolve this contradiction, generally falling on one side or another. (A very recent result found something in between, but in a way that, unhelpfully, didn’t agree with either side.) Gravitational lensing time delay measurements might be a way to resolve the problem, because with the VRO, the number of these systems we can use will go from a few to hundreds. Gravitational wave measurements from instruments like LIGO (discussed in Chapter 7) could give us insight here too, and in the next decade or so might reach the precision necessary to finally settle the question.

THE VIEW FROM LEFT FIELD